Angles of elevation and depression

1 σ or $\angle CBD$

2 $x = 53.13^\circ$

3 $x = 37.23^\circ$

4 $\theta = 52.52^\circ$

5 $h = 19\text{ m}$

6 $\theta = 41.81^\circ$

7 $x = 182$

8 $d = 229\text{ m}$

9 $\theta = 23.9$

Applications

10

a $x = 455.84\,\text{m}$ **b** $425.84\,\text{m}$

11

a $h = 127.38\,\text{m}$ **b** $129.1\,\text{m}$

12 $d = 1022.58$

13 $h = 17\text{ m}$

14 $d = 10.86$

15 $\theta = 45.97$

16

a $AB = 11.86\text{ m}$ **b** $AC = 16.24\text{ m}$ **c** $BC = 4.4\text{ m}$

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17

a $AC = 8663$ m **b** $BC = 6975$ m **c** 1688 m

18

a $x = 29.26^\circ$ **b** $y = 4.61^\circ$ **c** $z = 18.76^\circ$ **d** 100 cm

19

a $h = 9.37$ m **b** $\theta = 25^\circ$

20

a 61 m **b** 1601 m

21

a $CD = 22.65$ m **b** $AB = 18.29$ m

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